

GASTROINTESTINAL SYSTEM

NCLEX STUDY GUIDE

Assessment • Disorders • Medications • Surgical Care

1. Assessment and Upper GI Disorders

ABDOMINAL EXAM Use I-A-Pe-Pa: inspection, auscultation, percussion, then palpation.

GERD	Priority / management
Assessment	Heartburn, epigastric pain, regurgitation, dyspepsia and hypersalivation. New dysphagia may indicate stenosis/aspiration risk.
Risk factors	Caffeine, chocolate, alcohol, citrus, tomato, smoking, peppermint and drugs that reduce LES tone.
Interventions	Low-fat/high-fiber meals, avoid food/fluids near bedtime, elevate head 6-8 inches, avoid tight clothing and triggers.
Barrett esophagus	Premalignant change requiring follow-up; it is not itself esophageal cancer.

Gastritis	Peptic ulcer disease
H. pylori, NSAIDs/ASA, alcohol, smoking and irritants. Chronic gastritis may cause pernicious anemia requiring lifelong B12.	Gastric pain occurs 30-60 min after meals and food worsens it; hematemesis common. Duodenal pain occurs 1.5-3 hr after meals/at night and improves with food; melena common.

EMERGENCY Hypotension/tachycardia suggest hemorrhage. Sudden pain relief, rigid abdomen or pain radiating to the left shoulder may indicate perforation/peritonitis.

2. GI Medications and H. pylori

Class	Key points
Antacids	Aluminum: constipation; magnesium: diarrhea; calcium: constipation. Separate from other drugs.
H2 blockers	Famotidine; cimetidine has many interactions and may cause confusion in older adults.
Sucralfate	Protective coating; give on an empty stomach before meals.
PPIs	Omeprazole, pantoprazole, lansoprazole, esomeprazole; risks include C. difficile, fracture and kidney effects.
Misoprostol	Prevents NSAID ulcers; diarrhea/cramping; contraindicated in pregnancy.
H. pylori therapy	PPI plus antibiotics; quadruple therapy may add bismuth and metronidazole/tetracycline.

3. Gastric Surgery and Dumping Syndrome

Procedure	Connection
Billroth I	Partial gastrectomy with stomach-to-duodenum anastomosis.
Billroth II	Stomach-to-jejunum anastomosis.
Gastrectomy/bariatric surgery	Monitor early dumping syndrome and later B12 deficiency/pernicious anemia.

- Dumping occurs about 30 minutes after eating: cramping, borborygmi, diarrhea, dizziness, sweating, weakness and tachycardia.
- Small frequent meals, high protein, moderate/high fat as ordered, low simple carbohydrate; avoid fluids with meals and lie down about 30 minutes afterward.

- Avoid metoclopramide because it accelerates gastric emptying; lifelong B12 may be required after major gastric resection.

4. Viral Hepatitis

Type	Transmission / priorities
A	Fecal-oral; contaminated food/water/shellfish. Vaccine available; immune globulin after exposure when indicated.
B	Blood/body fluids, sexual/perinatal. Screen pregnancy; vaccine and HBIG for exposed/newborn clients according to weight and maternal status.
C	Blood exposure, injection drugs and older transfusions; no vaccine.
D	Occurs only with hepatitis B coinfection; prevent through HBV vaccination.
E	Fecal-oral/waterborne; high maternal/fetal mortality in pregnancy.

Stage	Findings
Preicteric	Flu-like symptoms, fatigue, anorexia, nausea/vomiting and elevated enzymes.
Icteric	Jaundice, clay stool, dark urine, pruritus and increased bilirubin.
Posticteric	Convalescence; jaundice/pain resolve and urine/stool normalize.

5. Acute Abdomen



Abdominal quadrants and common pain locations

Location	Common concern
RUQ	Gallbladder or liver
LUQ/left shoulder	Spleen or peritoneal irritation
Mid-epigastric to back	Pancreatitis; also evaluate cardiac/obstetric emergencies
RLQ	Appendicitis
LLQ	Diverticulitis

Location	Common concern
Flank to groin	Renal stone

Appendicitis

- Pain begins periumbilical/epigastric then localizes to McBurney point in RLQ; anorexia, nausea/vomiting, fever and rebound tenderness.
- Dunphy: pain with cough; Rovsing: RLQ pain with LLQ pressure; psoas/obturator signs may be positive.
- NPO, IV fluids, antibiotics, ice not heat, and no laxatives/enemas. Sudden relief may signal rupture/peritonitis.

Pancreatitis

- Severe epigastric pain radiating to back, worse supine/eating and relieved by fetal position; nausea/vomiting. Cullen and Grey Turner signs indicate bleeding.
- Lipase is most specific and remains elevated longer; monitor calcium, glucose, fluid status and ileus.
- NPO, gastric decompression when ordered, fluids/electrolytes, analgesia, pancreatic enzymes with food for chronic disease and insulin as needed.

6. Gallbladder Disease

- Cholecystitis: RUQ pain radiating to right scapula after fatty food, nausea/vomiting and positive Murphy sign.
- NPO during acute symptoms, NG decompression/IV fluids as ordered, analgesic/antispasmodic and antiemetic.
- After laparoscopic surgery, shoulder/free-air pain is expected; ambulation helps. Maintain low-fat small frequent meals.
- T-tube drainage is initially green-yellow/brown; keep bag below gallbladder, record output, and report absent/excessive drainage. Clamp around meals only as ordered.

7. Cirrhosis and Portal Hypertension

- Early: hepatomegaly, RUQ pain, weight loss, weakness and anorexia. Later: jaundice, spider angiomas, palmar erythema, caput medusae, edema/ascites and bleeding.
- Hepatic encephalopathy: confusion, lethargy, asterixis, writing difficulty, seizure/coma. Reduce ammonia with lactulose and antibiotics as ordered.
- Ascites: sodium/fluid restriction and potassium-sparing diuretic such as spironolactone; monitor electrolytes and daily weight.
- Avoid alcohol and hepatotoxic drugs. Give vitamins, iron/folate and vitamin K as indicated.

VARICES Hematemesis/melena with hypotension and tachycardia is an emergency. With a Sengstaken-Blakemore tube, label lumens and keep scissors at bedside for airway compromise according to protocol.

8. Inflammatory Bowel Disease

Ulcerative colitis	Crohn disease
Continuous colon/rectum disease; 16-20 bloody/purulent stools, tenesmus, anemia and high colorectal-cancer risk. Toxic megacolon is an emergency.	Skip lesions anywhere mouth-to-anus, commonly terminal ileum; cobblestone mucosa, 4-6 soft stools with mucus, steatorrhea, malnutrition, fistula and abscess.
• During acute flare: NPO/bowel rest, IV fluids/TPN as ordered, low-fiber high-protein/iron diet when eating, avoid dairy/gas-forming foods, caffeine, alcohol and smoking. • Treatments may include metronidazole/ciprofloxacin, sulfasalazine, steroids, azathioprine/cyclosporine and biologics such as infliximab/adalimumab. Monitor infection and avoid live vaccines with immunosuppression.	

9. Ostomy Care

Ostomy	Location/output/irrigation
Ileostomy	RLQ, liquid output; never irrigate; high dehydration, electrolyte and skin-breakdown risk.
Ascending colostomy	RUQ, liquid output; do not irrigate.
Descending/sigmoid	LLQ, formed stool; scheduled irrigation may be used after healing.

- Healthy stoma is pink/red and initially swollen. Report pale, purple, dark blue or black tissue.
- Cut appliance about 1/8 inch larger than stoma; empty pouch when one-third to one-half full.
- After surgery: NPO/NG until bowel function returns, advance diet gradually and encourage early participation in care.
- For irrigation, use tepid water, container at shoulder height, stop for cramps or forceful return; never irrigate an ileostomy.

10. Diverticular Disease

Diverticulosis	Diverticulitis
Outpouchings without inflammation; high-fiber diet helps prevent constipation/inflammation.	LLQ pain, fever, distention and possible blood. Acute care: NPO/bowel rest, IV fluids, metronidazole/ciprofloxacin and avoid seeds/fiber until resolved.

FINAL NCLEX FOCUS Prioritize GI bleeding, perforation/peritonitis, severe dehydration, obstruction, toxic megacolon, encephalopathy and acute changes in stoma color or output.